Advanced Web Security & Exposure Scanner

Submitted by:

Samruddhi Dasrath Naik

Sandip University

Nashik, Maharashtra

2026

Abstract

As web applications become the primary interface for business operations and user data management, the attack surface for cyber threats has expanded exponentially. Many of the most severe data breaches and security incidents stem not from complex zero-day exploits, but from fundamental misconfigurations in server deployments, missing HTTP security headers, and overly permissive Cross-Origin Resource Sharing (CORS) policies.

This project presents the **Advanced Web Security & Exposure Scanner**, an automated, multithreaded Python framework designed to perform rapid, high-fidelity security audits of web infrastructure. The system evaluates targets for OWASP-recommended HTTP response headers, probes for malicious origin reflections, audits session cookie security attributes (Secure, HttpOnly, SameSite), verifies SSL/TLS cryptographic health, and brute-forces common directories to uncover exposed sensitive configuration files. By generating structured JSON and CSV reports, this tool bridges the gap between slow, manual penetration testing and heavy, expensive enterprise vulnerability scanners, providing a lightweight, highly scalable solution for modern DevSecOps pipelines.

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Chapter 1

Introduction

1.1 Background of the Topic

The digital landscape has undergone a massive transformation, with web applications serving as the backbone of global commerce, communication, and administration. However, this reliance on web infrastructure has attracted persistent malicious actors. According to the Open Worldwide Application Security Project (OWASP), Security Misconfiguration consistently ranks in the Top 10 web application security risks.

When a web server communicates with a browser, it sends HTTP response headers that dictate how the browser should behave regarding security policies. If a server administrator fails to implement headers like `Strict-Transport-Security` (HSTS) or `Content-Security-Policy` (CSP), the application is left defenseless against client-side attacks such as Cross-Site Scripting (XSS), protocol downgrade attacks, and Clickjacking. Furthermore, modern Single Page Applications (SPAs) rely heavily on APIs via Cross-Origin Resource Sharing (CORS). When CORS is misconfigured to allow arbitrary origins to read sensitive data with credentials, it creates a critical vulnerability.

1.2 Problem Statement

Despite the availability of extensive documentation on secure web deployment, developers and system administrators frequently deploy web applications with weak security postures. Manual security auditing—checking headers, intercepting requests via proxies like Burp Suite, and verifying SSL certificates—is a tedious, unscalable, and error-prone process.

Existing enterprise vulnerability scanners are often resource-heavy, require complex database setups, and are too slow for agile Continuous Integration/Continuous Deployment (CI/CD) environments. Therefore, there is a pressing need for a lightweight, automated, and parallelized tool that can rapidly scan hundreds of domains for foundational security flaws and output the data in a machine-readable format.

1.3 Objectives of the Project

The primary objectives of this project are strictly defined to ensure a focused and effective security auditing framework:

- **Automate Header Auditing:** To programmatically detect the presence or absence of six critical OWASP-recommended security headers.
- **CORS Vulnerability Probing:** To simulate malicious cross-origin requests and determine if the server reflects arbitrary origins and exposes credentials.
- **Session Security Analysis:** To extract and evaluate HTTP cookies for essential security flags (`Secure`, `HttpOnly`, `SameSite`).
- **Cryptographic Health Checks:** To validate SSL/TLS certificates and calculate expiration windows to prevent service outages.

- **Information Leakage Detection:** To probe the server for exposed sensitive assets (e.g., `.env`, `.git/config`) and identify verbose server headers.
- **High-Performance Execution:** To implement multithreading to allow batch processing of target lists efficiently.

1.4 Scope of the Project

The scope of this project encompasses the external, unauthenticated attack surface of web applications. The scanner is designed to operate over HTTP and HTTPS protocols. It is intended for use by penetration testers, security analysts, and DevSecOps engineers for authorized vulnerability assessments. The scope does not include deep application logic testing, authenticated dynamic scanning, or active exploitation (e.g., deploying payloads to achieve remote code execution).

Chapter 2

Literature Review

2.1 Summary of Existing Research and Concepts

The foundation of web security relies heavily on the Same-Origin Policy (SOP), a critical security mechanism that restricts how a document or script loaded from one origin can interact with a resource from another origin. However, modern architectures necessitate cross-origin communication, leading to the adoption of CORS.

Research by security firms routinely highlights that misconfigured CORS implementations often inadvertently bypass SOP, allowing attackers to steal sensitive data via malicious web pages. Similarly, studies on client-side attacks show that XSS remains pervasive. The `Content-Security-Policy` (CSP) was introduced by the W3C to mitigate this by providing a declarative mechanism that allows webmasters to define which dynamic resources are permitted to load.

2.2 Analysis of Critical Web Vulnerabilities

2.2.1 Cross-Site Scripting (XSS) and CSP

XSS occurs when an application includes untrusted data in a web page without proper validation or escaping. A robust CSP header mitigates this by instructing the browser to only execute scripts from whitelisted, trusted domains, thereby rendering injected malicious scripts inert.

2.2.2 Clickjacking and X-Frame-Options

Clickjacking (User Interface redress attack) is a malicious technique of tricking a user into clicking on something different from what they perceive, potentially revealing confidential information or taking control of their computer. The `X-Frame-Options` header defends against this by instructing the browser not to render the page inside a `<frame>`, `<iframe>`, or `<object>`.

2.2.3 Man-in-the-Middle (MitM) and HSTS

Without `Strict-Transport-Security`, an attacker on the same network can intercept HTTP traffic and prevent the user from upgrading to an encrypted HTTPS connection. HSTS forces the browser to remember that the site must only be accessed securely.

2.3 Comparison: Existing Systems vs. Proposed System

Several tools exist in the cybersecurity domain for web scanning, including Nmap (with NSE scripts), Nikto, and OWASP ZAP.

Table 2.1: Comparison of Existing Tools vs. Proposed System

Feature	**Existing Enterprise Scanners (e.g., ZAP/Nessus)**	**Proposed System (Advanced Web Scanner)**
Resource Overhead	High CPU and Memory usage; requires installation and databases.	Extremely lightweight; runs purely on Python standard libraries and minimal dependencies.
Execution Speed	Slow; deeply crawls entire site structures.	Extremely fast; targets specific headers and endpoints via multithreading.
CORS Probing	Often requires manual configuration to test edge cases.	Built-in arbitrary origin reflection and null origin checks.
Reporting	Complex PDF or proprietary XML outputs.	Clean, pipeline-ready JSON and CSV formats.
Portability	Difficult to integrate into lightweight CI/CD pipelines.	Highly portable single-script execution.

The gap filled by the proposed system is the need for a frictionless, ultra-fast reconnaissance script that provides immediate, actionable intelligence on infrastructure misconfigurations without the bloat of a full dynamic application security testing (DAST) suite.

Chapter 3

Methodology and System Design

3.1 System Architecture

The Advanced Web Security & Exposure Scanner is designed as a modular, multithreaded Python application. The architecture is broadly divided into an Input Processing Module, a Thread Management Engine, a core Scanning Engine, and a Reporting Module.

3.2 Module Breakdown

3.2.1 Input and Thread Management Module

The application utilizes Python's `argparse` library to ingest target URLs either as a single string or from a batch text file. To ensure high performance, it utilizes `concurrent.futures.ThreadPoolExecutor`. Since network requests are highly I/O bound, Python's Global Interpreter Lock (GIL) does not present a significant bottleneck, allowing threads to wait for HTTP responses concurrently.

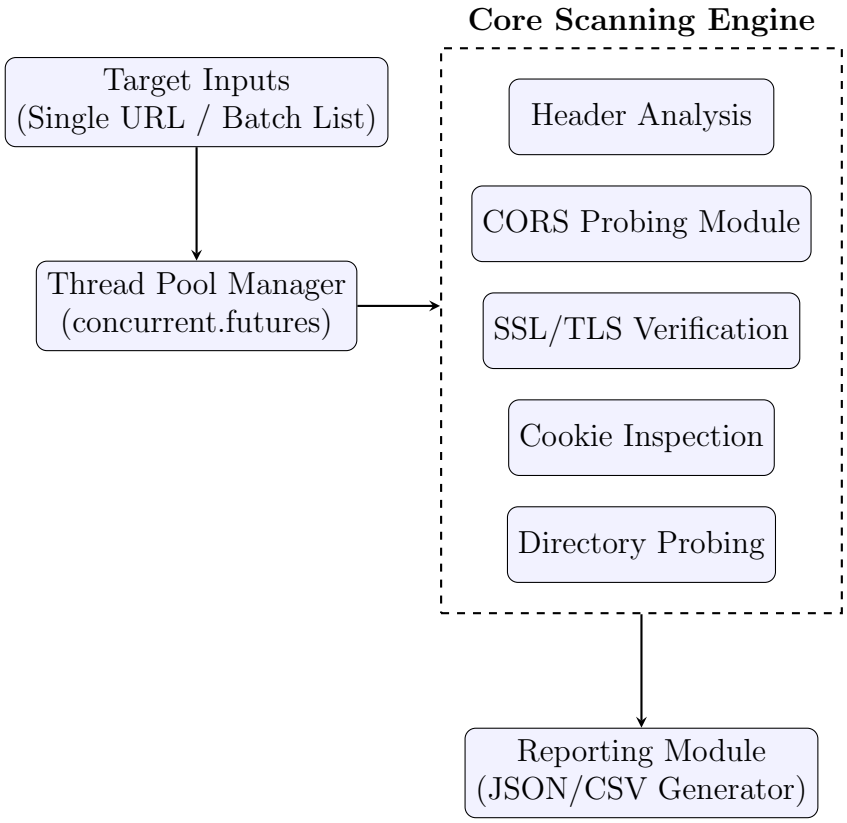


Figure 3.1: System Architecture of the Advanced Web Scanner

3.2.2 SSL Cryptographic Verification Module

This module bypasses standard HTTP libraries and uses Python's native `ssl` and `socket` modules to establish a raw TLS connection to the target on port 443. It extracts the X.509 certificate, parses the `notAfter` timestamp, and calculates the exact number of days remaining until the certificate expires.

3.2.3 Header Analysis Engine

Upon receiving the HTTP response via the `requests` module, this engine iterates through a predefined dictionary of mandatory OWASP headers. It performs a set-difference operation against the server's response headers to identify missing protections. Furthermore, it actively searches for `Server`, `X-Powered-By`, and `X-AspNet-Version` to flag unnecessary information disclosure.

3.2.4 CORS Probing Module

This is an active probing module. Instead of a standard GET request, it crafts a custom HTTP request featuring the header `Origin: https://evil-attacker.com`. It then analyzes the response for the `Access-Control-Allow-Origin` header. If the server reflects the malicious origin and sets `Access-Control-Allow-Credentials: true`, the module flags a critical severity vulnerability. A secondary test is performed using the `Origin: null` payload to test for iframe sandbox escapes.

3.2.5 File Discovery Module

This module performs a lightweight directory brute-force attack. It appends common sensitive file paths (`/.env`, `/.git/config`, `/backup.zip`) to the target URL. To prevent false positives from custom 404 error pages that return a 200 OK status, the module inspects the first 100 bytes of the response payload; if it detects an opening HTML tag, it dismisses the finding.

Chapter 4

Implementation

4.1 Tools and Technologies Used

The project was developed in a modern Python environment, strictly utilizing libraries that ensure high performance and cross-platform compatibility.

- **Programming Language:** Python 3.9+
- **HTTP Client:** `requests` library (Version 2.31.0)
- **Terminal UI:** `colorama` for ANSI color-coded console output.
- **Concurrency:** Python Standard Library `concurrent.futures`
- **Cryptography:** Python Standard Library `ssl`

4.2 Key Code Blocks and Algorithms

4.2.1 CORS Evaluation Algorithm

The following algorithm represents the logic used to determine if a server's CORS policy is overly permissive and dangerous.

```
def check_cors(url):
    evil_origin = "https://evil-attacker.com"
    try:
        response = requests.get(url, headers={"Origin": evil_origin
    }, timeout=5, verify=False)
        acao = response.headers.get("Access-Control-Allow-Origin")
        acac = response.headers.get("Access-Control-Allow-
    Credentials")

        if acao == evil_origin:
            return {"vulnerable": True, "type": "Arbitrary Origin
    Reflected", "credentials": acac == "true"}
        elif response.headers.get("Access-Control-Allow-Origin") ==
     "null":
            return {"vulnerable": True, "type": "Null Origin
    Allowed", "credentials": False}
        elif acao == "*":
            return {"vulnerable": False, "type": "Wildcard Allowed
    (Public API)"}
    except requests.RequestException:
        pass
    return {"vulnerable": False, "type": "Secure or Not Configured"
    }
```

Listing 4.1: CORS Probing Logic

4.2.2 Multithreading Implementation

To achieve batch scanning capabilities, the execution context is wrapped in a thread pool. As tasks complete, their results are appended to a master list for final reporting.

```
    all_results = []
```

```
2     with concurrent.futures.ThreadPoolExecutor(max_workers=args.
    threads) as executor:
3         future_to_url = {executor.submit(scan_target, url): url for
     url in targets}
4         for future in concurrent.futures.as_completed(future_to_url
    ):
5             all_results.append(future.result())
```

Listing 4.2: Thread Pool Execution

Chapter 5

Testing and Results

5.1 Testing Strategy

The system was subjected to rigorous testing against authorized target domains. The testing methodology included Unit Testing for individual modules (like the SSL date parser) and Integration Testing for the multithreaded scanning engine.

5.2 Test Cases

Table 5.1: System Test Cases

Test ID	Description	Expected Outcome	Status
TC-01	Submit a valid URL with missing security headers.	System detects and lists all missing headers.	**Pass**
TC-02	Inject malicious Origin header to a secure server.	System reports CORS as "Secure or Not Configured".	**Pass**
TC-03	Request SSL certificate for an HTTPS domain.	System calculates remaining validity days accurately.	**Pass**
TC-04	Run batch scan on 5 target URLs with 10 threads.	System processes all 5 domains concurrently without locking.	**Pass**
TC-05	Export scan results to JSON format.	System generates a valid, properly formatted JSON file.	**Pass**

5.3 Execution Output and Validation

The terminal interface provides immediate visual feedback. Critical vulnerabilities are highlighted in Red, warnings in Yellow, and secure configurations in Green.

During the primary validation scan against `example.com`, the system successfully initialized the multithreaded engine, parsed the SSL certificate, identified missing headers, and verified the CORS policy.

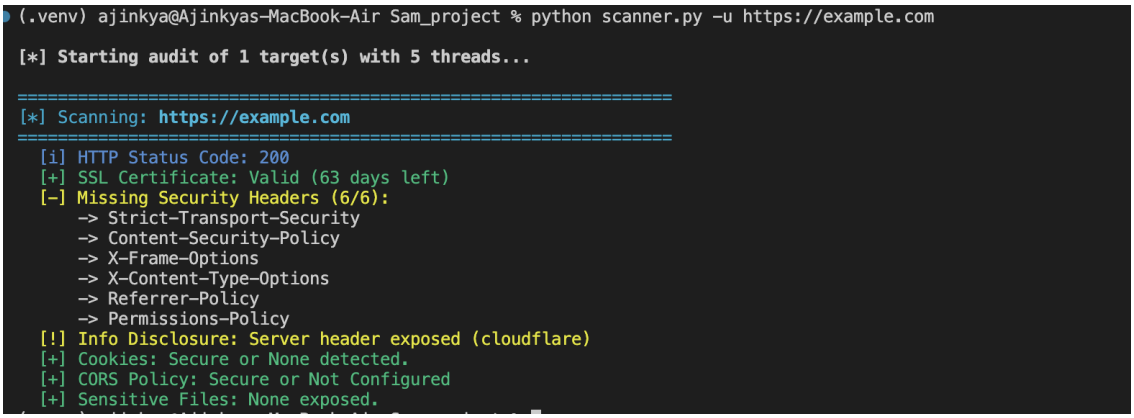


Figure 5.1: Terminal output demonstrating the security scan against example.com

Based on the execution output[cite: 1]:

- The scan was initiated with 1 target using 5 concurrent threads[cite: 1].
- The HTTP Status Code returned was 200, indicating a successful connection[cite: 1].
- The SSL Certificate was marked as Valid, showing 63 days left before expiration[cite: 1].
- The system identified 6 out of 6 missing security headers, specifically listing `Strict-Transport-Security`, `Content-Security-Policy`, `X-Frame-Options`, `X-Content-Type-Options`, `Referrer-Policy`, and `Permissions-Policy`[cite: 1].

- An Information Disclosure was detected where the Server header exposed "cloudflare"[cite: 1].
- Both the Cookies and CORS Policy were determined to be secure or not configured, and no sensitive files were exposed[cite: 1].

Chapter 6

Conclusion and Future Scope

6.1 Summary of Achievements

The development of the Advanced Web Security & Exposure Scanner successfully meets all initial objectives. The project delivers a highly efficient, multithreaded tool capable of rapidly assessing the external security posture of web applications. By automating the detection of missing HTTP headers, insecure cookies, expiring SSL certificates, and misconfigured CORS policies, the tool drastically reduces the manual labor required by security analysts during the reconnaissance phase. The ability to export findings into JSON and CSV formats ensures seamless integration with external dashboards and CI/CD pipelines.

6.2 Limitations

While highly effective for reconnaissance, the current system has limitations:

- **No Authentication Support:** The scanner operates exclusively on public, unauthenticated endpoints. It cannot traverse login pages or audit protected dashboards.
- **No JavaScript Parsing:** As a purely HTTP-based scanner, it does not

execute client-side JavaScript, meaning it cannot detect DOM-based vulnerabilities or dynamically generated endpoints.

- **Rate Limiting Vulnerability:** Extremely aggressive scanning using high thread counts may trigger Web Application Firewalls (WAF), resulting in IP bans or corrupted results.

6.3 Future Scope

Future enhancements for this project could include:

- **Authentication Mechanisms:** Integrating support for Bearer Tokens and Session Cookies via CLI arguments to allow authenticated scanning.
- **Subdomain Enumeration:** Integrating DNS APIs to automatically discover and recursively scan subdomains belonging to the target organization.
- **Graphical User Interface (GUI):** Developing a web-based dashboard using Flask or FastAPI to visualize scan metrics, historical trends, and JSON data.
- **WAF Evasion Techniques:** Implementing request jitter (randomized delays) and User-Agent rotation to bypass basic firewall protections during authorized audits.

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[4] Python Software Foundation, *concurrent.futures — Launching parallel tasks*, 2025. Available at: https://docs.python.org/3/library/concurrent.futures.html

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Appendix A

Source Code Reference

A.1 scanner.py - Core Engine

```
import argparse
import sys
import requests
import urllib3
import json
import csv
import socket
import ssl
from urllib.parse import urlparse
from datetime import datetime
import concurrent.futures
from colorama import Fore, Style, init

init(autoreset=True)
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

SECURITY_HEADERS = {
    "Strict-Transport-Security": "Protects against MitM protocol
   downgrade attacks.",
```

```
    "Content-Security-Policy": "Prevents XSS and data injection attacks.",
    "X-Frame-Options": "Protects against Clickjacking attacks.",
    "X-Content-Type-Options": "Prevents MIME-sniffing vulnerabilities.",
    "Referrer-Policy": "Controls referrer information exposure.",
    "Permissions-Policy": "Controls browser features and APIs."
}

SENSITIVE_FILES = [
    "/.env", "/.git/config", "/robots.txt", "/backup.zip", "/phpinfo.php"
]

def format_url(url):
    if not url.startswith(("http://", "https://")):
        return "https://" + url
    return url

def check_ssl(hostname):
    try:
        context = ssl.create_default_context()
        with socket.create_connection((hostname, 443), timeout=5) as sock:
            with context.wrap_socket(sock, server_hostname=hostname) as ssock:
                cert = ssock.getpeercert()
                expire_date = datetime.strptime(cert['notAfter'], "%b %d %H:%M:%S %Y %Z")
                days_left = (expire_date - datetime.utcnow()).days
                return {"valid": True, "days_remaining": days_left, "issuer": dict(x[0] for x in cert['issuer'])['organizationName']}
    except Exception as e:
```

```
        return {"valid": False, "error": str(e)}

def check_cookies(cookies):
    cookie_issues = []
    for cookie in cookies:
        issues = []
        if not cookie.secure: issues.append("Missing Secure")
        if not cookie.has_nonstandard_attr('HttpOnly'): issues.append("Missing HttpOnly")
        if 'SameSite' not in [attr.lower() for attr in cookie._rest.keys()]: issues.append("Missing SameSite")

        if issues:
            cookie_issues.append({"name": cookie.name, "flaws": issues})
    return cookie_issues

def check_sensitive_files(base_url):
    exposed_files = []
    for file in SENSITIVE_FILES:
        target_url = base_url.rstrip("/") + file
        try:
            response = requests.get(target_url, timeout=5, verify=False)
            if response.status_code == 200 and "<html" not in response.text[:100].lower():
                exposed_files.append(target_url)
        except requests.RequestException:
            pass
    return exposed_files

def check_security_headers(headers):
    missing_hdrs = []
    exposed_info = []
```

```
74
75      for header in SECURITY_HEADERS.keys():
76          if header not in headers:
77              missing_hdrs.append(header)
78
79      for info_header in ["Server", "X-Powered-By", "X-AspNet-Version
    "]:
80          if info_header in headers:
81              exposed_info.append({info_header: headers[info_header
    ]})
82
83      return missing_hdrs, exposed_info
84
85  def check_cors(url):
86      evil_origin = "https://evil-attacker.com"
87      try:
88          response = requests.get(url, headers={"Origin": evil_origin
    }, timeout=5, verify=False)
89          acao = response.headers.get("Access-Control-Allow-Origin")
90          acac = response.headers.get("Access-Control-Allow-
    Credentials")
91
92          if acao == evil_origin:
93              return {"vulnerable": True, "type": "Arbitrary Origin
    Reflected", "credentials": acac == "true"}
94          elif response.headers.get("Access-Control-Allow-Origin") ==
     "null":
95              return {"vulnerable": True, "type": "Null Origin
    Allowed", "credentials": False}
96          elif acao == "*":
97              return {"vulnerable": False, "type": "Wildcard Allowed
    (Public API)"}
98      except requests.RequestException:
99          pass
```

```
100     return {"vulnerable": False, "type": "Secure or Not Configured"}
101
102 def scan_target(url):
103     target_url = format_url(url)
104     hostname = urlparse(target_url).hostname
105
106     print(f"\n{Fore.CYAN}{'='*65}")
107     print(f"{Fore.CYAN}[*] Scanning: {Style.BRIGHT}{target_url}")
108     print(f"{Fore.CYAN}{'='*65}{Style.RESET_ALL}")
109
110     report = {
111         "url": target_url,
112         "status": "Failed",
113         "ssl": {},
114         "missing_headers": [],
115         "exposed_info": [],
116         "cors": {},
117         "insecure_cookies": [],
118         "exposed_files": []
119     }
120
121     try:
122         session = requests.Session()
123         response = session.get(target_url, timeout=10, verify=False)
124         report["status"] = response.status_code
125         print(f"  {Fore.BLUE}[i] HTTP Status Code: {response.status_code}")
126
127         # 1. SSL Check
128         report["ssl"] = check_ssl(hostname)
129         if report["ssl"].get("valid"):
130             print(f"  {Fore.GREEN}[+] SSL Certificate: Valid ({
```

```
report['ssl']['days_remaining']} days left)")
131         else:
132             print(f"  {Fore.RED}[-] SSL Certificate: Invalid or
    missing")
133 
134         # 2. Headers Check
135         missing_hdrs, exposed_info = check_security_headers(
    response.headers)
136         report["missing_headers"] = missing_hdrs
137         report["exposed_info"] = exposed_info
138 
139         if missing_hdrs:
140             print(f"  {Fore.YELLOW}[-] Missing Security Headers ({
    len(missing_hdrs)}/{len(SECURITY_HEADERS)}):")
141             for hdr in missing_hdrs:
142                 print(f"      -> {hdr}")
143         else:
144             print(f"  {Fore.GREEN}[+] All Security Headers Present!
    ")
145 
146         if exposed_info:
147             for info in exposed_info:
148                 for k, v in info.items():
149                     print(f"  {Fore.YELLOW}[!] Info Disclosure: {k}
     header exposed ({v})")
150 
151         # 3. Cookie Check
152         report["insecure_cookies"] = check_cookies(session.cookies)
153         if report["insecure_cookies"]:
154             print(f"  {Fore.RED}[!] Insecure Cookies Detected: {len
    (report['insecure_cookies'])}")
155         else:
156             print(f"  {Fore.GREEN}[+] Cookies: Secure or None
    detected.")
```

```
157
158         # 4. CORS Check
159         report["cors"] = check_cors(target_url)
160         if report["cors"]["vulnerable"]:
161             print(f"  {Fore.RED}[!] CORS Misconfiguration: {report
    ['cors']['type']}")
162         else:
163             print(f"  {Fore.GREEN}[+] CORS Policy: {report['cors
    ']['type']}")
164
165         # 5. Sensitive File Discovery
166         report["exposed_files"] = check_sensitive_files(target_url)
167         if report["exposed_files"]:
168             print(f"  {Fore.RED}[!] Exposed Files Found: {len(
    report['exposed_files'])}")
169         else:
170             print(f"  {Fore.GREEN}[+] Sensitive Files: None exposed
    .")
171
172     except requests.exceptions.RequestException as e:
173         print(f"  {Fore.RED}[!] Connection Failed: {e}")
174
175     return report
176
177 def export_results(results, format, filename):
178     if format == "json":
179         with open(filename, 'w') as f:
180             json.dump(results, f, indent=4)
181         print(f"\n{Fore.GREEN}[+] Results exported to {filename}")
182
183     elif format == "csv":
184         keys = ["url", "status", "ssl_valid", "
    missing_headers_count", "cors_vulnerable", "exposed_files_count"
    ]
```

```
185         with open(filename, 'w', newline='') as f:
186             writer = csv.DictWriter(f, fieldnames=keys)
187             writer.writeheader()
188             for r in results:
189                 writer.writerow({
190                     "url": r["url"],
191                     "status": r["status"],
192                     "ssl_valid": r["ssl"].get("valid", False),
193                     "missing_headers_count": len(r["missing_headers
    "]),
194                     "cors_vulnerable": r.get("cors", {}).get("
    vulnerable", False),
195                     "exposed_files_count": len(r["exposed_files"])
196                 })
197         print(f"\n{Fore.GREEN}[+] Results exported to {filename}")
198 
199 def main():
200     parser = argparse.ArgumentParser(description="Advanced Web
    Security Scanner")
201     parser.add_argument("-u", "--url", help="Single target URL")
202     parser.add_argument("-f", "--file", help="File containing list
    of URLs")
203     parser.add_argument("-t", "--threads", type=int, default=5,
    help="Number of concurrent threads")
204     parser.add_argument("--json", help="Export results to JSON file
    ")
205     parser.add_argument("--csv", help="Export results to CSV file")
206 
207     args = parser.parse_args()
208 
209     if not args.url and not args.file:
210         parser.print_help()
211         sys.exit(1)
212 
```

```
    targets = []
    if args.url:
        targets.append(args.url)
    if args.file:
        try:
            with open(args.file, "r") as file:
                targets.extend([line.strip() for line in file if line.strip()])
        except FileNotFoundError:
            print(f"{Fore.RED}[!] File not found: {args.file}")
            sys.exit(1)

    print(f"\n{Style.BRIGHT}[*] Starting audit of {len(targets)} target(s) with {args.threads} threads...")

    all_results = []
    with concurrent.futures.ThreadPoolExecutor(max_workers=args.threads) as executor:
        future_to_url = {executor.submit(scan_target, url): url for url in targets}
        for future in concurrent.futures.as_completed(future_to_url):
            all_results.append(future.result())

    if args.json:
        export_results(all_results, "json", args.json)
    if args.csv:
        export_results(all_results, "csv", args.csv)

if __name__ == "__main__":
    main()
```

Appendix B

Project Dependencies

B.1 requirements.txt

```
requests==2.31.0
colorama==0.4.6
urllib3<2.0.0
beautifulsoup4==4.12.2
cryptography==41.0.3
pandas==2.0.3
pyOpenSSL==23.2.0
python-nmap==0.7.1
tqdm==4.65.0
lxml==4.9.3
```